

# Yash Rawal

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## Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms, hierarchical RAG architectures, and automated intelligence pipelines across 120+ organizations. Specialized in Python, Model Context Protocol (MCP), multi-agentic orchestration (LangChain, LangGraph), and healthcare NLP. Active contributor to Spring AI, bridging generative AI and model fine-tuning with resilient cloud deployments (Azure, AWS).

## Skills

**Languages & Backend:** Python, SQL, PostgreSQL, Java, FastAPI, Spring Boot, REST APIs

**AI & ML Engineering:** Agentic AI, Model Context Protocol (MCP), Multi-Agent Systems, Tool RAG, Healthcare NLP

**AI & ML Frameworks:** LangChain, LangGraph, Spring AI, PyTorch, vLLM, LlamaIndex, GROBID

**Platforms & Models:** Azure, AWS, Azure OpenAI, Amazon Bedrock, Meta Llama, Claude, Qwen

**Infrastructure & Security:** Docker, Kubernetes, Linux, CI/CD, LLM-as-Judge, Red Teaming, PII Sanitization

## Work Experience

**Bigscal Technologies Pvt. Ltd. (Client: TalenTeam Ltd., UK)**

**AI/ML Engineer**

Oct 2024 - Present

Surat, India

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants on cloud infrastructure (Azure, AWS).
- Architected and deployed multi-agent workflows with LangChain and LangGraph, utilizing **MCP Server Tool Routing (Tool RAG)** and A2A communication with integrated guardrails.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by 70%.
- Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
- Engineered an automated LLM evaluation and red-teaming control plane to log reasoning trajectories, scoring agent behaviors via LLM-as-a-judge and **RLHF preference datasets**.
- Fortified agentic workflows against OWASP LLM vulnerabilities with phantom-token context isolation, runtime tool telemetry, and **automated PII redaction**.

**Bigscal Technologies Pvt. Ltd. (Product: Healthray)**

**AI/ML Engineer Intern**

Apr 2024 - Sep 2024

Surat, India

- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR with **NeMo RNN-T and CTC decoding**.
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

## Projects

**Modular Agent Harness & Memory System ("Brain")** | Open Source

**Principal Creator & Architect** of an open-source plug-and-play agentic framework designed as modular building blocks for flexible enterprise workflows, implementing a formal ReAct loop with step budgets, vector tool routing, and dynamic MCP settings caching.

Java, Spring Boot, PostgreSQL, WebSockets

**Spring AI (spring-projects/spring-ai)** | Open Source

Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.

Java, Spring Boot, Python

**Mini Language Model (~85M Parameter Architecture)** | Personal Research

Pretrained an ~85M-parameter transformer to ~1.43 validation loss on TinyStories targeting **edge hardware deployment**; engineered FragmentStream chunked attention to eliminate  $O(T^2)$  VRAM overhead and custom Triton kernels with KV caching for low-latency inference.

Python, PyTorch, CUDA, Triton, Tesla P100/T4

## Education

**Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University**

2022 - 2024

## Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)

Microsoft

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Microsoft